POWERFUL TEACHING



UNLEASH THE SCIENCE OF LEARNING

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Introduction

Do you remember your first day teaching, with students eagerly (or not eagerly) awaiting your instruction? Perhaps you felt terrified. Elated. Cautious. Ready. Not ready. Almost ready?

In the days leading up to that first day, you may wonder, "What am I going to say? What am I going to teach? What are they going to think of me? What should I do when my lesson plans tank?"

But how often do you ask, "*How* am I going to teach?"

Let's travel back in time to 2006. Patrice was starting her twelfth year teaching and Pooja had just graduated from college with a degree in elementary education. A veteran teacher and a newly minted teacher. Strikingly, the critical question on their minds was identical: What works best in education?

By this point, Patrice had noticed that her students were remembering what they learned years *after* her class – much more than what they remembered from other classes. Patrice also noticed that most teaching strategies were based on anecdotes, passed down from teacher to teacher. Year after year, teachers would come and go, but ineffective teaching methods would stay. *How* could there be such a disconnect between what was working in Patrice's classroom and ineffective fads permeating many other classrooms?

A few years earlier, Pooja had taken a course in cognitive science, without really knowing what that meant. She was astounded to realize that there is rigorous research on how humans learn and remember. *How* could there be such a disconnect between Pooja's education classes on one side of campus and her cognitive science classes on the other side of campus – both all about learning, but one based on anecdotes and the other based on science?

Suddenly, just like pieces in a puzzle, *it all came together*. Researchers at Washington University in St. Louis received grant funding to conduct cognitive science research in classrooms. Up until this point, most research within this specific field took place in laboratory settings with college students, using fairly simple materials to examine how students learn and remember information. Now, with this new grant, the long-overdue opportunity to bridge the gap between learning in the classroom and learning in the lab had arrived.

On a sunny Wednesday in August, Patrice and Pooja met for the first time – just inside the school's entrance, near the cafeteria, next to the principal's office. Shaking hands and walking up the stairs to Patrice's classroom, the energy was unmistakable; Patrice would be one of the first K–12 teachers to partner with cognitive scientists in her classroom, and Pooja would be one of the first cognitive scientists to conduct full-time research in an authentic school setting. This initial meeting of the minds became a teacher-scientist bond that continues today.

This book is the culmination of our extensive collaboration to understand what works best in education. Enter *the science of learning*. What, exactly, is the "science of learning?" It's so simple, and yet counterintuitive. Why can we remember scenes from our favorite movies but struggle to remember our students' names? Why is it easy to learn some things and hard to learn other things? Why can we read a book and feel we learned a lot, only to find that we've forgotten most of it a year later? Why can we think back and remember our first day of teaching, but not our tenth day of teaching?

Cognitive scientists conduct research on all of these things – how we learn stories, names, facts, important events, unimportant events, and more. Research on the science of learning dates back more than 100 years. Fads feel like they've persisted for 100 years, too. Sometimes we run into fads during pre-service programs and professional development, and other times through word of mouth or online blogs. We've all had the experience of being excited about a "new" teaching idea, only to file the materials away for a rainy day. Rarely are these new strategies utilized. Professional development programs often feel like a waste of time.

So why do we keep reinventing the wheel with teaching strategies based on the "fad of the semester," when scientifically based strategies are waiting to be unleashed? We feel there are two main reasons:

1. The science of learning sits dormant in academic journals, rather than easily accessible in pre-service textbooks and professional development materials.
2. The science of learning has recently been featured in newspapers, blogs, and social media, but it's hard to know if these are trusted sources or simply people concocting more fads.

For these reasons (and more), teachers are given the impossible challenge of finding time to seek out good research, make sense of it, and apply it in classrooms. We frequently get asked, "Why haven't I heard about this research before? Why didn't I learn about this in my pre-service program or professional development?" It's because this research isn't accessible – literally and figuratively. There is valuable research on the science of learning out there, but it's sitting behind lock and key.

Until now. For more than a decade, we have developed a rare partnership of scientist-and-teacher, research-into-practice – one that is true to the rigors of cognitive science and also practical with today's teachers in mind. In *Powerful Teaching: Unleash the Science of Learning*, we decipher cognitive science research, illustrate how we have applied the science of learning in our own teaching, and provide evidence-based recommendations to empower educators to unleash the science of learning in their classrooms.

There is valuable research on the science of learning out there, sitting behind lock and key. Until now.

By drawing on empirical research by fellow cognitive scientists and practical strategies from educators around the world, *we focus on four powerful teaching strategies based on the science of learning*: retrieval practice, spaced practice, interleaving, and feedback-driven metacognition.

1. *Retrieval practice* boosts learning by pulling information *out* of students' heads (e.g., quizzes and flashcards), rather than cramming information into students heads (e.g., lectures). Retrieval practice is a no-stakes learning opportunity that increases student performance, beyond formative and summative assessments.
2. *Spaced practice* boosts learning by spreading lessons and retrieval opportunities out over time so learning is not

crammed all at once. By returning to content every so often, students' knowledge has had time to rest and be refreshed.

3. *Interleaving* boosts learning by mixing up closely related topics and encouraging discrimination. For example, learning increases when students practice addition, subtraction, multiplication, and division problems all mixed up, rather than one type of problem at a time.
4. *Feedback* boosts learning by providing the student the opportunity to know what they know, and know what they don't know. This increases students' *metacognition* or understanding about their own learning progress.

Critically, research demonstrates that these four powerful strategies:

- Raise student achievement by a letter grade, or even two – *from a C to an A*. Research we conducted in Patrice's classroom and additional classrooms demonstrated a consistent and reliable increase in students' grades, confidence, and engagement.
- Boost learning for diverse students and subject areas. Our book applies to all grade levels and disciplines (e.g., STEM, social studies, language arts, fine arts, special education, and foreign languages).
- Enhance higher-order learning and students' transfer of knowledge. We provide research and examples demonstrating that these strategies apply for basic fact knowledge, skill learning, and critical thinking.

When it comes to retrieval practice, spaced practice, interleaving, and feedback-driven metacognition, *the combination of being research-based and classroom-proven is paramount*. The rigor of science gives us confidence that these strategies aren't fads, and successful classroom implementation gives us

confidence that these strategies work in the real world, not just in the laboratory. In addition, the better you understand the research behind the strategies, the more effectively you can adapt them in *your* classroom – and you know your classroom best.

By this point, you may be thinking, "These strategies sound great, but they probably take a lot of time and effort to use them." Actually, they don't! The research-based strategies we describe in this book can be implemented *in less than a minute without additional prep or grading time*. In addition, as teachers, we already use many of these strategies: retrieving what we know, spacing it out over time, mixing it up, and giving our students feedback. The difference is that we share how to use these evidence-based strategies *purposefully, intentionally, and frequently*.

We start this book by sharing the research behind retrieval practice and why it's foundational for learning. Next, we share specific, actionable strategies to implement retrieval practice in the classroom. We continue with research and classroom strategies for spacing, interleaving, and metacognition, followed by recommendations on how to overcome potential challenges, engage in conversations with students and parents, and lead evidence-driven professional development at the school, district and university level.

We have even more activities, templates, and downloads available on our website, www.powerfulteaching.org. Lastly, we have intentionally embedded retrieval, spacing, interleaving, and feedback throughout this book. In other words, we practice what we preach, we model these strategies in each chapter, and we boost *your* learning from this book!

With *Powerful Teaching: Unleash the Science of Learning*, you will:

- Develop a deep understanding of powerful teaching strategies based on the science of learning, whether you are a past, present, or future educator.

- Go behind the scenes and explore key findings from cognitive science research.
- Gain insight into how scientifically-based strategies are effectively implemented in a variety of academic settings without additional preparation, classroom, or grading time.
- Think critically about your current teaching practices and classroom environment from a research-based perspective.
- Develop tools to share the science of learning with students and parents, ensuring success inside and outside the classroom.
- Identify next steps to transform teaching and unleash the science of learning in your classroom.

As educators, we must look forward – using evidence-based strategies – to propel and guide student learning. We must stop driving instruction with anecdotes and fads, we must stop reinventing the wheel, and we must stop riding in circles when it comes to classroom instruction.

Instead, it's our responsibility to *ask for evidence*. Evidence for which strategies are effective, not just which strategies are popular. Evidence for why we teach the way we teach, not just because we've always taught that way.

Most importantly, after reading this book, we hope that on the first day of the school year, you won't be asking, "What should I teach?" Instead, you'll be asking, "*How* should I teach? *How* can I turn my teaching into *powerful teaching*?"

The science of learning exists. *Now, it's time to unleash it*.



Chapter 2

Build a Foundation with Retrieval Practice

We'd like to start this chapter with a few questions. Think about them carefully and write down your answers:

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What did you have for breakfast yesterday?[1]

At what age did King Tut become a pharaoh?[2]

What was your favorite vacation?[3]

What's one thing you've learned from this book so far?[4]

In 2017, Pooja asked some of these questions during an interview with Jennifer Gonzalez for the podcast Cult of Pedagogy. Here's what Jenn came up with:

> Pooja: What did you have for breakfast yesterday?
>
> Jenn: I have the exact same thing every day: spinach, mushrooms, eggs, and oatmeal.
>
> Pooja: Wow. That's much nicer than my coffee! Okay, here's another question: In what year did King Tut become a pharaoh?
>
> Jenn: Oh my gosh. Do you know, I used to know all about King Tut. When I was in fifth-grade my teacher was obsessed. I have no idea. It was a BC, right?
>
> Pooja: Yep, yep.
>
> Jenn: 500 BC? Am I anywhere near?
>
> Pooja: Do you remember how old he was?
>
> Jenn: Oh, he was a kid, wasn't he? Wasn't he like 10 or something?
>
> Pooja: Yes, he was 9 or 10 years old!
>
> Jenn: Right!

What's with these harebrained questions? Well, they help us talk about something that's tough to pinpoint – learning. In this example, you may have learned something at some point, but forgot it (e.g., King Tut's age), or you can simply think back and bring something to mind even if you didn't intentionally learn it (e.g., your breakfast yesterday). You may have a fond memory for something you haven't thought about in a while (e.g., your favorite vacation) or you can probably remember something you recently learned with a little bit of effort (e.g., one thing from this book).

Sound confusing? It's because it is. *Learning is complex and messy*. It's not something we can touch, and it's really hard to define. You might even say that the learning process looks more like a blob

than a flowchart. Reading Pooja and Jenn's conversation gives us initial insight into what learning and remembering "look" like. Now, get ready to dive in and explore the blob of learning!

THE THREE STAGES OF LEARNING AND THE CRITICAL IMPORTANCE OF RETRIEVAL

In Chapter 1, we talked about three stages of learning: encoding, storage, and retrieval.

- Encoding is when we initially learn something.
- Storage is how long something is retained in mind.
- Retrieval is when we access information and bring it to mind.

Using the three stages of learning and thinking about your responses to the four questions we just asked, complete the following table.

	Did you intentionally try to encode this information?	How long have you had this information stored?	How many times have you retrieved this information?
What you ate for breakfast			
King Tut's age			
Your favorite vacation			
One thing about this book			

When it comes to the classroom, where do we focus our activities and teaching practices – encoding, storage, or retrieval? Do you think of these stages as linear – like the arrow figure in Chapter 1 – or do you envision a meandering path across a map?

As teachers, we typically think of learning as getting something in through encoding; we hope what we teach gets stored in our students' heads; and we determine whether our students learned something by having them retrieve during an assessment.

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Where do you focus your teaching: encoding, storage, or retrieval?
Do you focus on one stage more than another stage?

In other words, we typically focus on getting information *into* students' heads. On the contrary, one of the most robust findings from cognitive science research is the importance of getting information *out* of students' heads. Based on a century of research, in order to transform learning, we must focus on getting information *out* – a strategy called *retrieval practice*. In fact, research demonstrates that retrieval practice is more potent than other techniques commonly used by teachers and students, such as lecturing, re-reading, or taking notes.

Retrieval and the process of getting information *out* is so simple, intuitive, and flexible. For instance, even though the four questions we just asked are very different, they are all examples of retrieval – we're reaching back and pulling information out about breakfast, kings, vacations, and books. Formative assessments and exit tickets are another intuitive form of retrieval. But why do we still default to traditional encoding-assessment methods, like "I lecture, you write a paper" and "I teach a lesson, you take a test?"

When we think about learning, we typically focus on getting information into students' heads. What if, instead, we focus on getting information out of students' heads?

In this chapter, we highlight the critical importance of retrieval for learning, not just encoding. More specifically, we focus on our first Power Tool, *retrieval practice*, and we present conclusive scientific evidence that it significantly benefits learning.

THE POWER OF RETRIEVAL PRACTICE IN PATRICE'S CLASSROOM

After a long day of teaching, I (Patrice) stopped at a Thai restaurant for take-out dinner. While I was waiting, two former students I hadn't seen in six years came in. The second they saw me, they came over and exclaimed: "Mrs. Bain!" They started a rapid-fire litany of knowledge learned in my class. Each time one student retrieved a fact, it would trigger more retrieval from the other student. They continued to rattle off information; I felt like I was watching a ping-pong volley of knowledge. And their facts were correct! What was exciting to me was that we hadn't expected to see each other; there was no preparation on their part. Yet, how was it possible, after a delay of six years, for them to pull out so much information?

Was this an isolated incident? Absolutely not! As another example, I received a lovely note from a parent this year. Her daughters were in my class a few years previously. The family had taken a trip to Europe over the summer and the mother was astonished by the facts and stories the girls told her at each historical site. Each time the girls were asked how they knew this information, their replies were always: "Oh, we learned that in Mrs. Bain's class." In fact, rarely does a school year go by when I have *not* been contacted by surprised parents – and students – with similar stories of delight in remembering details from years past.

Students often ask, "Why do I remember so much from your class, but I don't remember that much from my other classes?"

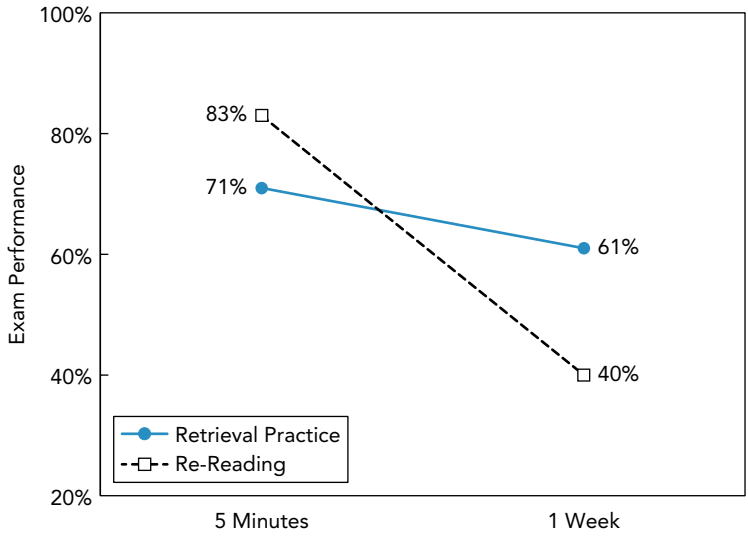
Why do teachers get the deer-in-the-headlights look from students

when they don't remember information from year to year? Or semester to semester? Or even week to week? At the same time, why do my students retain so much information? It's because of the strategies I use. In this case, *retrieval practice*. Not only does research overwhelmingly support retrieval practice, I have used it for over 10 years in my classroom. The success rate of my students is consistent. Retrieval practice works.

THE SCIENCE BEHIND POWER TOOL #1: RETRIEVAL PRACTICE IN THE LABORATORY

First, how do we know that the process of retrieval improves learning? Consider a landmark study conducted in 2006 by our colleagues Henry L. Roediger, III ("Roddy") and Jeffrey Karpicke.[5] In this simple laboratory study, college students at Washington University in St. Louis read brief passages about topics like sea otters and the sun. After reading the passages once, some students were instructed to

Figure 2.1 College students remember information after re-reading in the *short term*, but retain significantly more after retrieval practice over the *long term*.



re-read them over and over again, whereas other students retrieved information by writing down everything they could remember.

Importantly, Roddy and Jeff measured students' performance on an exam five minutes later and also one week later using the same "write down everything you can remember" method. What they found was pretty stunning (Figure 2.1):

- After five minutes, student performance in the re-reading condition was much higher (83%) compared to the retrieval condition (71%).
- After one week, this pattern completely switched – student performance in the retrieval condition (61%) was now much higher than performance in the re-reading condition (40%).

Consider it this way: In the span of one week, when students re-read passages, they forgot more than *half* of what they once knew! On the other hand, when students engaged in a simple, quick retrieval method, they forgot a little bit (which is expected), but they forgot much *less* over the one-week delay.

In addition to examining short-term and long-term learning, Roddy and Jeff also asked college students, "How much will you remember in one week?" They asked this question for passages that were re-read and also for passages that were followed by retrieval practice.

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Do you think college students predicted they will remember more after re-reading or after retrieval practice? Why?

On a 0–7 point scale, students said they'd remember more after re-reading (a rating of 4.8) than after retrieval (a rating of 4.0) – the opposite from the actual results. Time and time again,

cognitive scientists find this same pattern or *illusion of fluency*, where students think they'll remember something well after re-reading, but that's actually when they forget the most.[6]

This study by Roddy and Jeff is one simple example that when we deliberately recall information, it forces us to pull our knowledge out and examine what we know. We won't go into all the research here, but thousands of laboratory and neuroscience studies demonstrate that retrieval practice significantly increases learning.[7]

In typical laboratory experiments, college students do the following:

1. They study a set of material (e.g., foreign language vocabulary words, reading passages).
2. They engage in retrieval practice (e.g., via free-recall or multiple-choice quizzes).
3. They complete a final exam immediately or after a delay (e.g., ranging from minutes, to hours, to days).

Consistently, when students engage in retrieval activities that bring knowledge to mind, researchers see a dramatic increase in both short-term and long-term learning.[8]

Quick question: Who was the fourth president of the United States? Instantly, a plausible answer may have jumped to mind, only to realize you had to mentally struggle to think of the actual answer. As we discussed in Chapter 1, it's precisely this struggle or *desirable difficulty* during retrieval that improves learning. Now that you have struggled with this question, check the answer at the end of this chapter.[9] Thanks to our first Power Tool, you'll probably remember the name of the fourth president much better after retrieval practice than if we simply told you the name!

THE SCIENCE BEHIND POWER TOOL #1: RETRIEVAL PRACTICE IN AUTHENTIC CLASSROOMS

We know that learning about sea otters and basic trivia facts can be fun, and these simplistic materials work well in controlled laboratory environments. But what about a chapter in a real Advanced Placement (AP) Biology book or formulas from a college engineering course? Does retrieval practice improve long-term learning in authentic classroom settings for real classroom materials?

Yes!

Research in classrooms has demonstrated that retrieval practice improves learning for:

- Diverse student populations (e.g., K–12 students to medical school students)
- Subject areas (e.g., introductory history to CPR skills)
- Time delays (e.g., ranging from weeks to months)
- Developmental stages (e.g., preschool, young adults, and older adults)[10]

Even with fire alarms, snow days, assemblies, absences, and the day-to-day chaos in classrooms, *scientists have raised grades from a C to an A* for thousands of students, simply with retrieval practice. How? Let's dive into our bread and butter: research with real students, real materials, in real classrooms.

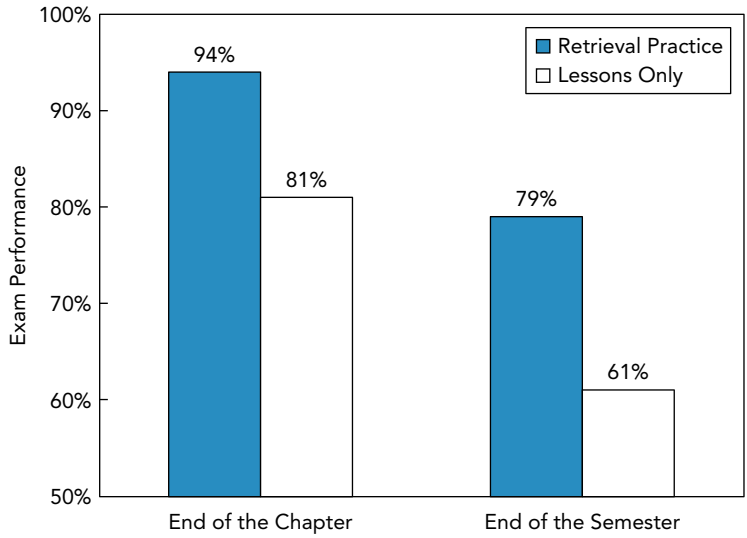
Research has shown that retrieval practice can increase students' grades from a C to an A.

In one of the first studies to explore retrieval-based learning in an authentic classroom setting, students in Patrice's sixth-grade social studies class received three brief low-stakes quizzes via clicker remotes. For a chapter on Ancient Egypt, for example, all information was covered during Patrice's lesson, but half of the chapter material was presented on

the clicker quizzes and half the chapter material wasn't quizzed. In other words, students engaged in retrieval practice for only part of the chapter material, but all of the material was covered during Patrice's lessons.

The team of researchers from Washington University in St. Louis (Pooja, Roddy, Mark McDaniel, and Kathleen McDermott) found that student performance was significantly greater for quizzed material than for nonquizzed material (94% vs. 81%) on a chapter test. Even at the end of the semester (approximately one to two months after Patrice's lessons), student performance was still significantly greater for quizzed material than nonquizzed material *by an entire letter grade* (79% vs. 67%; see Figure 2.2).[11]

Figure 2.2 Patrice's sixth-grade students had significantly higher exam performance when her lessons included retrieval practice (three brief quizzes) compared to lessons without retrieval practice.



By this point, we hope you're thinking, "Of course, students do better when they're quizzed versus not quizzed. They're seeing the material more often." Now you're thinking like a scientist! In a follow-up experiment, researchers added a condition where students could read material via the clicker software, but they couldn't click in. In other words, if simply seeing the information is as effective as retrieval practice, then scores should be the same in both conditions. Sure enough, on the end-of-chapter exam, quizzes again raised students' grades by an entire letter grade (91% vs. 81%). Even something as simple as three quizzes made a huge impact on student learning.

Consider one study by Keith Lyle and Nicole Crawford in college statistics classes at the University of Louisville.[12] The classes were split into two conditions:

- One class engaged in retrieval practice via short-answer questions at the end of each lecture.
- Another class continued with "business as usual" lectures. They didn't receive any short-answer questions and the lectures continued a little longer to fill the same amount of time as the retrieval practice condition.

When Keith and Nicole looked at four exams throughout the semester, performance for students in the retrieval practice condition was nearly a letter grade higher (86%) compared to students in the lecture-only condition (78%).

Scientists have shown that even medical students benefit from retrieval practice.[13] In one study by Doug Larsen and colleagues, medical students taking a course on neurology participated in three initial learning conditions:

- Students listened to a lecture and then practiced identifying neurology conditions with a standardized patient (an actor who is trained to engage in conversation as a patient).

- Students listened to a lecture and then took a written quiz on key topics.
- Students listened to a lecture and studied a review sheet of key topics.

After six months, performance on a written test was significantly greater when students had initially engaged in the two retrieval conditions (61%) than when they only studied a review sheet (48%). When the final test assessed students' performance on a new standardized patient, initial retrieval practice with a standardized patient (the first condition) led to the greatest performance after six months (59% vs. 49% vs. 43%, respectively).

While this is just the tip of the scientific iceberg so far, in each chapter we'll continue to present research that demonstrates that such a small adjustment during teaching – like a quick quiz – can make a *big* difference for student learning.

HOW TO DEFINE RETRIEVAL PRACTICE IN *YOUR* CLASSROOM

Throughout this book, when we talk about retrieval practice, it is described as bringing information to mind or pulling information out. We retrieve every day, all the time – from our favorite movie scenes to whether we turned off the kitchen stove (we hope you did!).

And even though we retrieve information nearly every second of every day, it's an abstract concept that can be difficult for us to describe when we talk with students, parents, and other teachers. From our experience, everyone defines retrieval practice in different ways. And that's not a bad thing! But how can we be equipped to describe it to others so we're all on the same page?

Doug Lemov, author of *Teach Like a Champion 2.0*, defines retrieval practice as follows:

> Retrieval practice occurs when learners recall and apply multiple examples of previously learned knowledge or skills after a period of forgetting.[14]

Doug breaks down his definition as follows:

- Retrieval practice is relevant to skills (hitting a baseball, solving for the perimeter of an octagon) and to knowledge (historical dates, formulas, vocabulary).
- In the classroom, retrieval practice is intentional, beyond simply asking questions.
- Retrieval practice requires at least a short delay after something has been learned because once you've started to forget you have to work harder to remember.

In a survey we conducted with teachers around the world in 2018, we asked how they define retrieval practice. Here are a few of our favorites:

- Having to strain your memory in order to do something previously learned
- Reaching into the recesses of your brain to grab knowledge that's stored there
- Recalling what you've been learning, without the pressure of having your results evaluated
- Digging for information and bringing it up to the learning surface
- Actively generating knowledge from memory as opposed to simply reading

Whether you define retrieval practice as recalling examples, digging into your brain, or pulling information out, we encourage you to think about (and retrieve!) how you define retrieval

practice for yourself, for students, and also for other teachers. It's complex, it's powerful, and it's also just a fun part of everyday life.

Power Up

How do you define retrieval practice?

TEN BENEFITS OF RETRIEVAL PRACTICE

In Chapter 3, we dig into specific ways to incorporate retrieval practice in your teaching. Before we talk about the *how* of retrieval practice, we'd like to first present the *why*. In other words, *why* should you consider using this Power Tool at all?

Here are 10 (of the many) important benefits of retrieval practice, adapted from a list by Roddy, Adam Putnam, and Megan (Smith) Sumeracki.[15] Retrieval practice:

1. Improves students' learning and retention of information over the long term
2. Increases students' higher-order thinking and transfer of knowledge
3. Identifies students' gaps in knowledge, which provides formative assessment for teachers and students
4. Increases students' metacognition and awareness of their own learning
5. Increases students' engagement and attention in class
6. Increases students' use of effective study strategies outside of class
7. Increases students' advance preparation for class
8. Improves students' mental organization of knowledge
9. Increases students' learning of related information that isn't initially retrieved

10. Increases students' learning in the future by blocking interfering information

More than 100 years of research demonstrates that learning can be enhanced, boosted, and transformed with something as simple as getting information out.[16] In fact, retrieval practice yields even *more* benefits for learning when it's used as the basis on which we add more Power Tools we'll be talking about in this book: spacing, interleaving, and feedback-driven metacognition. Without retrieval practice, additional Power Tools improve learning on their own; but *with* retrieval practice, these strategies make a huge impact on student learning.

RETRIEVAL PRACTICE BOOSTS HIGHER-ORDER THINKING AND TRANSFER OF KNOWLEDGE

By now, you may be wondering, "Retrieval practice sounds great, but it still sounds pretty basic." Over the years, we've had the same concerns, too! Fortunately, in the lab and in the classroom, research demonstrates that retrieval practice boosts students' higher-order thinking, application of knowledge, and skills like writing and math. We'll provide research on how retrieval practice boosts more than just memorization, but before we do, let's take a look at complex learning in Patrice's classroom.

Higher-Order Thinking in Patrice's Classroom

Learning facts is crucial and it is the first step. But learning is much more than the simple regurgitation of facts. Learning is also higher-order thinking; it's the synthesis and evaluation of information. For example, if you ask a student, "What color is the sky?" you can anticipate that from pre-K on, the answer will be blue. That is a terrific answer for early elementary grades.

However, we can't stop there. From upper elementary grades on, we should ask, "*Why* is the sky blue?" Immediately, students

are forced to pull out or retrieve information and synthesize it into an answer. The higher the grade level, the more information can be retrieved and synthesized.

One of the ways I (Patrice) encourage higher-order thinking in my classroom is by using the acronym *HOT*. In fact, when students write an essay or answer an essential question, I emphasize: *Let's make this so HOT it sizzles!*

It is vital to take learning beyond lower-level thinking. Knowing facts readily assists students in the ability to form opinions and answer essential questions. But I strive for higher-order thinking, where students ponder, synthesize facts, and analyze how the puzzle fits together.

It's important to know that students earn high grades in my class. Students who earn Cs and lower in other classes usually earn As and Bs in mine. Is it because my class is easier than others? *Quite the contrary*. Because of the Power Tools I use, students not only learn facts – they are able to engage in higher-order thinking, too.

Research Demonstrates Retrieval Practice Boosts Students' Knowledge *and* Higher-Order Thinking

Retrieval practice boosts students' higher-order thinking, application of knowledge, and skills like writing and math.

Of course, the development of students' higher-order thinking is a critical component of education. From both an educational perspective and a scientific perspective, it is of practical interest to develop robust strategies that improve higher-order thinking and learning. What, exactly, is higher-order thinking and learning? While there are few agreed-on definitions, higher-order thinking is frequently classified using *Bloom's Taxonomy*, first published in 1956, followed by a revision to the taxonomy in

2001.[17] Based on Bloom's Taxonomy (frequently depicted as a pyramid), higher-order thinking is considered to comprise the apply, analyze, evaluate, and create categories closer to the top of the pyramid. On the other hand, recognition, memory, and comprehension fall under the remember and understand categories near the bottom of the pyramid.

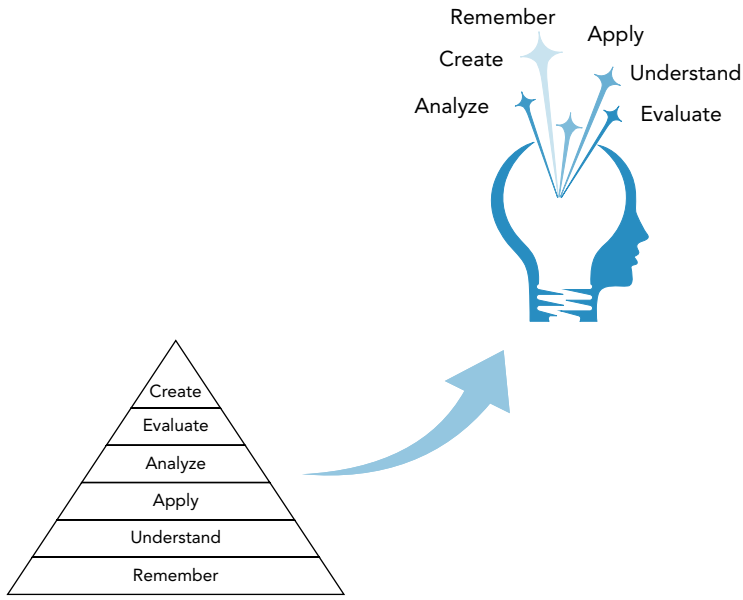
Does retrieval practice improve more than skills at the bottom of the pyramid? Yes! In one study in an eighth-grade science classroom, the same team from Washington University in St. Louis mentioned earlier found that when retrieval practice included both fact questions and application questions, students' performance on the final exam was higher than if they didn't receive retrieval practice at all.[18] Importantly, the application questions had the biggest boost on performance on the final exam with complex questions. Similarly, in an experiment conducted in Patrice's classroom as part of Pooja's dissertation, a mix of basic concept questions and higher-order questions during retrieval practice produced the highest exam performance – more than basic questions alone.[19]

Research also demonstrates that retrieval practice improves students' learning of skills, like math and language arts. In one study led by Charles Kromann, researchers looked at how well medical students could perform CPR. After students received training, some students practiced CPR on a dummy for 10 minutes while other students didn't. When assessed on the key steps of CPR after two weeks, students who initially engaged in retrieval practice significantly outperformed students who didn't engage in any practice (83% vs. 73%).[20]

When it comes to retrieval practice, *how far up the pyramid can we move student learning*? If we want students to think on a higher-order level, then we should make sure our retrieval questions are basic *and* higher-order. It's shortsighted to think, "Gee, well, if I have students retrieve a vocabulary word, they should be

able to apply this in a higher-order example or a higher-order type of material." Based on research, *provide a mix of fact-based retrieval and higher-order retrieval* if that's the type of learning you want to see in your students.[21] In other words, move from *pyramid learning* to *powerful learning* in which all levels of thinking and learning are engaged during retrieval practice (Figure 2.3).

Figure 2.3 Move from *pyramid learning* to *powerful learning* by providing *a mix* of fact and higher-order retrieval practice.



Keep in mind that learning basic knowledge is a good thing! Can a surgeon perform surgeries well without an understanding of basic anatomy? Knowledge plays a critical role in providing a foundation for learning, problem solving, and building connections.[22] Emerging research suggests that learning is not unidirectional or a hierarchy; we don't need to necessarily learn facts before we can think critically, analyze, or engage in higher order

thinking. Instead, particularly in the context of retrieval practice, a bi-directional route and a mix of basic and complex knowledge can be more beneficial for students' higher-order thinking.

One additional note about knowledge: Recently, Pooja had the opportunity to video chat with high school students in Indiana. A few hours later, Pooja led a video chat with teachers at a high school in South Carolina. Within the same day, *both* high school students and high school faculty asked this: What does Google mean for learning, memory, and knowledge? Do we have to learn or teach basic facts?

Fascinating question! Of course, we've all had the experience of Googling something, perhaps without even attempting to retrieve it (e.g., how well can you remember how ice cream is made? Can't remember? Google it!). Because students have Google at their fingertips, they may not need to learn specific dates, locations, and facts to the same extent as decades ago. At the same time, students can't rely on Google for everything. Sticking with our surgeon analogy, you wouldn't want a surgeon Googling instructions in the middle of a challenging operation! As another example, how can a student understand the nuances of a complex political event without understanding the basics of the political parties involved?[23]

Scientist and professor Jason Finley, who specializes in interactions between memory and technology, put it this way:

> We don't really know all the implications of Google yet. We know that people are memorizing facts less and it's becoming more important to know how to *find* information than to have it all in your head.

Back to the bigger picture: Knowledge enhances higher-order thinking (and vice versa). With retrieval practice, include complex questions, not just fact questions. Anecdotally, we've heard from teachers who use retrieval practice to help students learn

Buddhism, sign language, and dressage (a performance discipline for riding horses). Talk about knowledge *and* higher-order thinking!

Research Demonstrates That Retrieval Practice Boosts Students' Transfer of Knowledge

Let's say a teacher teaches a lesson on the water cycle. A few days later, a student is walking outside. She notices that a puddle from the morning has vanished by the afternoon. How can the teacher enhance the student's understanding that evaporation has occurred? How can students successfully apply what they have learned from one lesson to another, from one class to another, or from school to the real world?

Transfer happens when students take something familiar *and apply it to something* unfamiliar.

Transfer of learning, or simply called transfer, is the application of concepts or information in new situations.[24] In other words, transfer happens when students take something *familiar* and apply it to something *unfamiliar*. In the same way, a fundamental goal is that student learning *inside* the classroom is applied *outside* the classroom – the hallmark of deep understanding and flexible knowledge.

Retrieval practice enhances students' transfer of learning, which can take many forms. For example, a type of *near transfer* could involve students learning how to calculate the Pythagorean theorem, followed by calculating the theorem with a new set of numbers. In contrast, *far transfer* would be when a medical student learns something from a textbook and then applies it when diagnosing patients (see Figure 2.4 for more examples).[25]

In a meta-analysis from 2018, scientists Steven Pan and Tim Rickard at the University of California, San Diego, found that retrieval practice produces more robust transfer than any other

Figure 2.4 Retrieval practice improves students' transfer of knowledge, ranging from near to far transfer and also transfer in a variety of settings.

	NEAR TRANSFER ←			→ FAR TRANSFER
Knowledge	Ancient Egypt in 1330 BC vs. 1325 BC	Ancient Egypt vs. Ancient China	Ancient Egypt vs. Modern United States	Ancient Egypt vs. Romantic Literature
Physical	Same classroom	Different classroom at the same school	Different schools	School vs. home
Time	During the same lesson	During the same day	Weeks or months later	Years later
Task	Algebraic calculation with new numbers	Algebraic calculation with novel diagrams	Algebraic calculation with word problems	Algebraic calculation with real world examples
Format	Same format as before	Multiple-choice vs. short answer	Multiple-choice vs. essay	Multiple-choice vs. projects

research-based strategy to date.[26] Why does retrieval practice boost transfer? Because it's a desirable difficulty. Students learn information in a deeper manner than when they simply review information. With this increased effort and broader engagement, students don't simply recall information; they have to use existing knowledge in new and different ways.

Here's the catch: While we might *know it when we see it*, transfer of learning can be *easier said than done*, too. For example, when students are asked, "On what day did the Allies invade Normandy?" (D-Day) and then later asked, "Where did the Allies invade on D-Day?" (Normandy), it's surprising that students struggle to transfer their knowledge; transfer from one specific part of a fact to another can be tough.

While we might know it when we see it, transfer of learning can be easier said than done.

Here's another situation where college students surprisingly struggle to transfer their knowledge. In the 1980s, researchers created a "fortress problem" and a "tumor problem." In the fortress problem, a military general wanted to capture a fortress in the center of a country. The problem was that large groups of soldiers could not travel on the same road to the fortress. The simple solution was that *soldiers had to travel in small groups, taking different roads to reach the fortress*.

In the tumor problem, a physician was trying to irradiate a tumor. The problem was that a single dose of radiation strong enough to destroy the tumor would also severely damage nearby tissue. Can you guess the solution? It might be obvious to you, but it wasn't to the college students! The vast majority of the college students (80%) failed to transfer and apply the fortress solution to the new tumor problem (note that this sample was small; only 15 students).[27]

Thankfully, not all is lost! Even though retrieval practice improves students' transfer, scientists have demonstrated there are two additional strategies to ensure success: *hints* and *feedback*. For the fortress and tumor problems, when students were given a helpful hint, nearly all of them (92%) generated the correct solution.

Why is transfer more challenging than we expect?

- Students may not *recognize* that their knowledge should be applied in a novel situation.
- Students may recognize that they need to apply their knowledge, but they can't *remember* the knowledge to be applied. If they can't remember it, they can't transfer it.
- Students may not correctly *apply* knowledge to the novel situation.

How can we reduce these challenges and ensure successful transfer?

- Hints help students recognize when and what to transfer.
- Retrieval practice helps students remember what to transfer.
- Feedback helps students apply their knowledge correctly.

When it comes to higher-order thinking, skill learning, and transfer of knowledge, retrieval practice is a key Power Tool. Combined with helpful hints and meaningful feedback (which we discuss in greater detail in Chapter 5), retrieval practice is for more than just memorization or basic fact learning – *much* more. Daniel Willingham, professor and author of *Why Don't Students Like School*, put it this way:

> A great deal has been written about the impact of retrieval practice on memory. That's because the effect is sizable, it has been replicated many times and it seems to lead not just to better memory, but deeper memory that supports transfer.

RETRIEVAL PRACTICE IS NOT THE SAME AS FORMATIVE OR SUMMATIVE ASSESSMENTS

You're probably familiar with these two forms of assessment:

Formative assessment is when we check on and monitor students' progress. Kind of like inserting a toothpick into a cake to see how it's doing while baking, formative assessment is a strategy to see if things are on track.

Summative assessment is when we discover what students have learned through various forms of measurement. This is when we get to celebrate accomplishments, perhaps like enjoying a cake after baking, while also noticing what could be done differently next time.

But where does retrieval practice fit in?

In keeping with our cake analogy, *retrieval practice* is the process of learning how to crack an egg, measure ingredients, and mix it all together (perhaps using a Power Blender!). This is when we embrace mistakes rather than emphasize perfection because challenges are a *good thing* for learning.

Consider an additional comparison to a file drawer:

- Retrieval practice: Organize the files so it's easier to access next time.
- Formative assessment: Peek into a file drawer and see what's inside.
- Summative assessment: Inventory all the files before moving to the next cabinet.

There are, of course, similarities and differences between retrieval practice, formative assessment, and summative assessment:

- *Similarity*. All three require retrieval! From the outside, it can look like one seamless process, and that's a good thing.
- *Difference*. Retrieval practice doesn't require any data collection, grading, or points. *Nothing* needs to be recorded in the gradebook. Retrieval practice is a *no-stakes opportunity* when students can experiment, be challenged, and improve over time.

Retrieval practice is a learning strategy, not an assessment strategy.

This is why retrieval practice does *not* equal assessment. While formative and summative assessment are valuable for *checking* learning, retrieval practice activities, like brief quizzes we'll discuss in Chapter 3, are valuable for *increasing* learning. Similarly, retrieval practice is *a learning strategy*, not an assessment strategy.

It's also crucial that we think of retrieval as a *good* thing for learning, not a bad thing that's only associated with assessment. We must foster a classroom culture of retrieval that's *positive, not negative*. By keeping retrieval practice no-stakes, we (and our students!) get away from thinking of retrieval as more tests and move toward retrieval as being *more learning*.

Get away from thinking of retrieval as more tests and move toward retrieval as being more learning.

Cognitive scientists used to refer to retrieval practice as the *testing effect* (and occasionally still do in the research literature). But you won't find the word *test* in this book very often. It's because retrieval and pulling information out is what dramatically improves learning. Retrieval practice can take many forms – including *Think-Pair-Share* and flashcards – not just tests.

Lastly, retrieval practice is *not* a call for more testing; *it's the opposite*. Retrieval practice is successful when we foster learning in a supportive, no-stakes environment that values oddly shaped cupcakes over a perfect wedding cake.

Power Up

We described three reasons why students struggle to transfer their knowledge. What is one of those reasons?

TAKE RETRIEVAL PRACTICE FROM YOUR CLASSROOM TO YOUR EVERYDAY LIFE

Now that you have a firm understanding of retrieval practice and the science behind it, pay attention to it; not only inside your classroom, but also in your daily life. When you start paying attention to your own retrieval habits, you'll be surprised at how often you bring information to mind *outside* the classroom.

Then, take it to the next level and start practicing retrieval for all sorts of things: where you parked your car, the name of a book you want to recommend to a colleague (hint!), and even details from your day you want to share with a loved one. Start incorporating retrieval practice into your everyday life and experience firsthand how desirable difficulties are a good thing for your *own* learning.

It's time to move from what we ate *yesterday* to transforming learning *tomorrow*. Get ready, get set – let's go!

NOTES

1. Yesterday, Pooja probably had coffee for breakfast, and Patrice likely had Greek yogurt and half a banana.
2. Don't cheat! Take a guess and write it down before checking the answer.
3. Pooja's favorite vacation was a two-week road trip through Uruguay. Patrice's favorite vacation was snorkeling around the US Virgin Islands.
4. We mean it! We know you're near the beginning of the book, but we hope you've learned at least one thing already. Pause and write down one thing you've learned so far!
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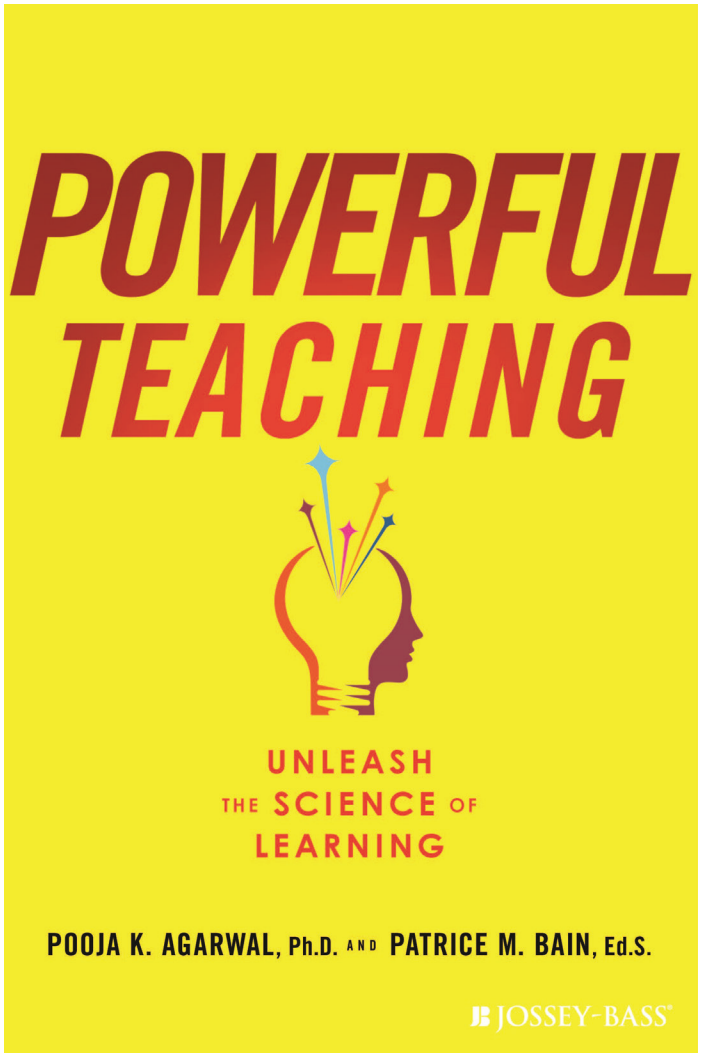
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